

Experiment 3: Heterogeneity at the Grokking Phase Boundary

Federated Learning Grokking Project

March 2026

1 Experiment 3: Heterogeneity at the Phase Boundary

1.1 Overview

Experiment 3 investigates whether data heterogeneity shifts the grokking phase boundary and, if so, through what mechanism. Building on the centralized phase boundary established in Experiment 1 ($\alpha_{\text{crit}} \approx 0.2$) and the federated baseline from Experiment 2 (FedAvg-IID matches centralized step-for-step), we now vary the *type* and *severity* of heterogeneity across clients.

The experiment has two sub-experiments:

- **Exp 3a** sweeps a continuous heterogeneity knob (Dirichlet concentration $\alpha_{\text{dir}} \in \{0.01, 0.1, 0.5, 1.0, 10.0, 1000.0\}$) across five training fractions $\alpha \in \{0.2, 0.25, 0.3, 0.35, 0.5\}$ at $K=10$ clients, with a $K=20$ validation subset.
- **Exp 3b** compares structured partitions—operand-based and target-based—against the IID baseline at the same α values.

All runs use $N=256$, SGD with $\eta=50$, $\lambda=0$, $E=5$ local epochs, full participation ($f=1$), and $S_{\text{max}}=50,000$ gradient steps. Each configuration is repeated over three seeds (42, 123, 456).

1.2 Exp 3a: Dirichlet Heterogeneity Sweep

1.2.1 Phase diagram

Figure 1 presents the grokking phase diagram as a function of training fraction α and Dirichlet concentration α_{dir} . The key observations are:

- **The phase boundary is unchanged.** At $\alpha=0.2$, grokking fails universally—across all six heterogeneity levels, no seed achieves $T_{\text{grok}} < \infty$. At $\alpha \geq 0.25$, all 30 configurations (5 α_{dir} values $\times$ 6 seeds) grok to 100% test accuracy. Heterogeneity does not shift α_{crit} .
- **Heterogeneity delays grokking, but only near the boundary.** At $\alpha=0.25$, extreme non-IID ($\alpha_{\text{dir}}=0.01$) increases T_{grok} from $26,725 \pm 2,048$ (IID) to $36,638 \pm 1,727$ —a 37% slowdown. At $\alpha=0.5$, the effect vanishes: T_{grok} varies by less than 1.3% across all α_{dir} values (7,638–7,738 steps).
- **The slowdown saturates quickly.** Most of the delay is concentrated between $\alpha_{\text{dir}}=1000$ and $\alpha_{\text{dir}}=0.1$; further increasing heterogeneity to $\alpha_{\text{dir}}=0.01$ adds only modest additional delay (Figure 2).

Table 1 summarizes the full $\alpha \times \alpha_{\text{dir}}$ grid. The monotonic increase in T_{grok} from right (IID) to left (non-IID) is visible at every α that groks, but the absolute magnitude of the effect shrinks rapidly as α increases.

Table 1: **Exp 3a: Full Dirichlet sweep** ($K=10$, mean $T_{\text{grok}} \pm \text{std}$ over 3 seeds). Slowdown column shows ratio vs. IID ($\alpha_{\text{dir}}=1000$).

α	$\alpha_{\text{dir}}=0.01$	0.1	0.5	1.0	10.0	1000 (IID)	Max slow.
0.20	∞	∞	∞	∞	∞	∞	—
0.25	36638 ± 1727	29497 ± 3366	27282 ± 2503	26910 ± 2189	26767 ± 1942	26725 ± 2048	$1.37\times$
0.30	14448 ± 831	13690 ± 595	13222 ± 493	13190 ± 513	13087 ± 437	13072 ± 449	$1.11\times$
0.35	10608 ± 377	10167 ± 238	9993 ± 205	9963 ± 193	9942 ± 203	9933 ± 194	$1.07\times$
0.50	7738 ± 105	7680 ± 126	7638 ± 118	7642 ± 120	7645 ± 113	7642 ± 120	$1.01\times$

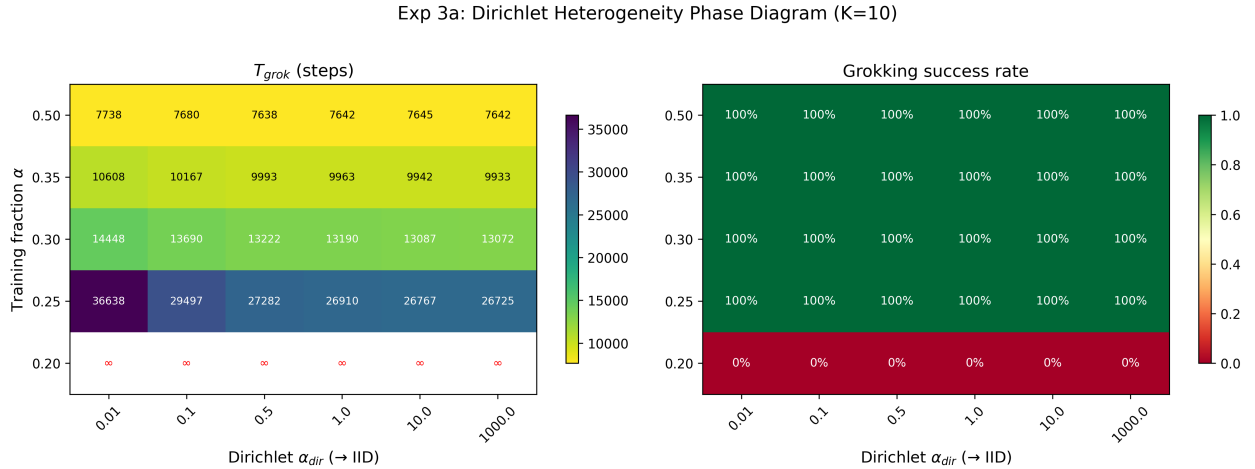


Figure 1: **Exp 3a phase diagram.** Left: mean T_{grok} (steps) as a function of α and α_{dir} ($K=10$). Right: grokking success rate (fraction of seeds that grok). Grokking succeeds at 100% for all $\alpha \geq 0.25$, regardless of heterogeneity. $\alpha=0.2$ universally fails (red ∞ markers).

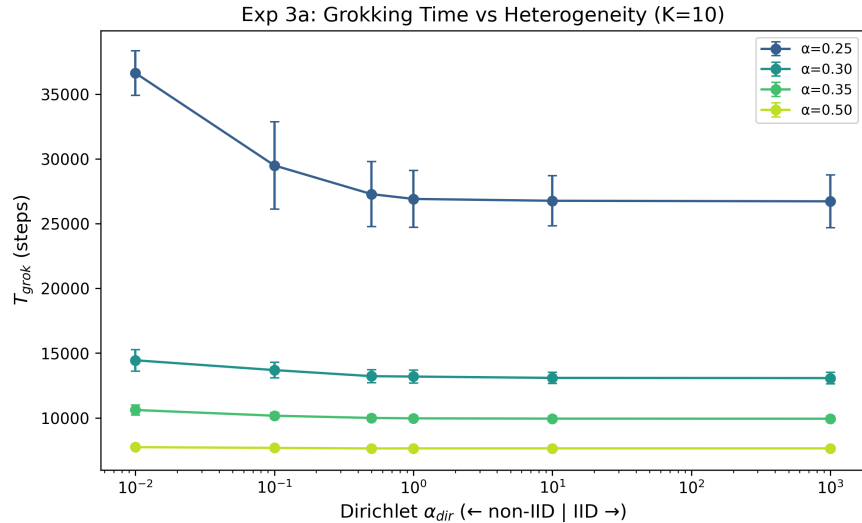


Figure 2: **Grokking time vs. heterogeneity.** T_{grok} (mean $\pm$ std over 3 seeds) as a function of α_{dir} for each training fraction. The effect of heterogeneity is largest near the phase boundary ($\alpha=0.25$) and negligible in the interior ($\alpha=0.5$). Note log scale on x -axis.

1.2.2 K -validation

Figure 3 compares $K=10$ and $K=20$ at three representative $(\alpha, \alpha_{\text{dir}})$ settings. Doubling the number of clients amplifies the heterogeneity effect modestly: at $\alpha=0.25$ with $\alpha_{\text{dir}}=0.1$, T_{grok} increases from 29,497 ($K=10$) to 34,977 ($K=20$), an 18.6% further slowdown. At $\alpha=0.5$, the $K=10$ and $K=20$ curves are indistinguishable. The qualitative pattern is preserved: more clients slightly amplifies the delay near the boundary but does not shift α_{crit} .

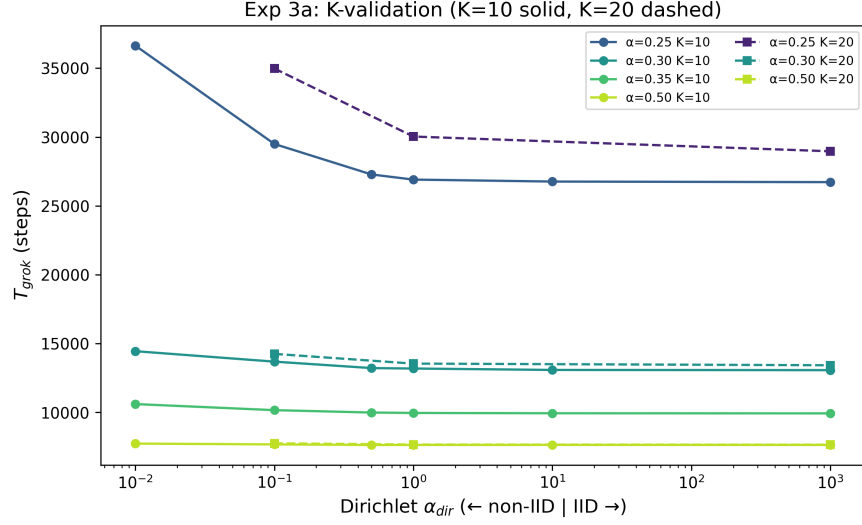


Figure 3: **K -validation.** $K=10$ (solid) vs. $K=20$ (dashed). Increasing K modestly amplifies heterogeneity-induced delay near the boundary but does not shift the phase boundary itself.

1.3 Exp 3b: Structured Partitions

1.3.1 Operand vs. target vs. IID

A central hypothesis motivating this experiment was that operand-based partitioning—which fragments the input space along the Fourier structure that the model must learn—would be more disruptive than Dirichlet quantity skew. The results **do not support this hypothesis** (Figure 4, Table 2).

- At $\alpha=0.5$, all three partitions yield statistically identical T_{grok} : IID = 7,642, operand = 7,597, target = 7,612.
- At $\alpha=0.25$ (near the boundary), operand partitioning is actually *slightly faster* than IID ($0.965\times$), while target partitioning is slightly slower ($1.114\times$).
- Both structured partitions fail at $\alpha=0.2$, matching the IID baseline—the phase boundary is unchanged.

The trajectory-level comparison (Figure 5) confirms that the learning dynamics (test accuracy and IPR curves) are nearly overlapping for all three partition types at every α .

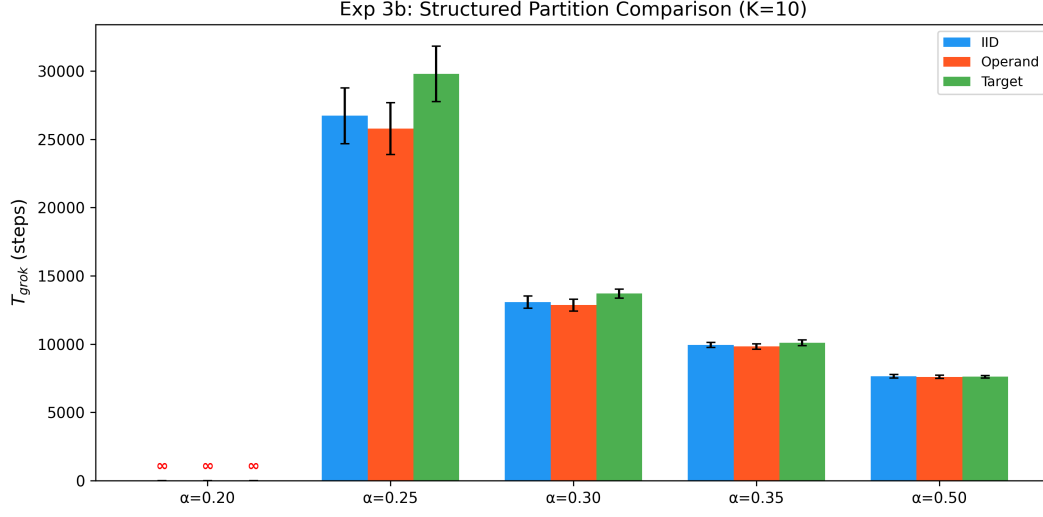


Figure 4: **Exp 3b: Structured partition comparison.** T_{grok} by partition type for each α ($K=10$). Operand and target partitions produce grokking times within noise of IID.

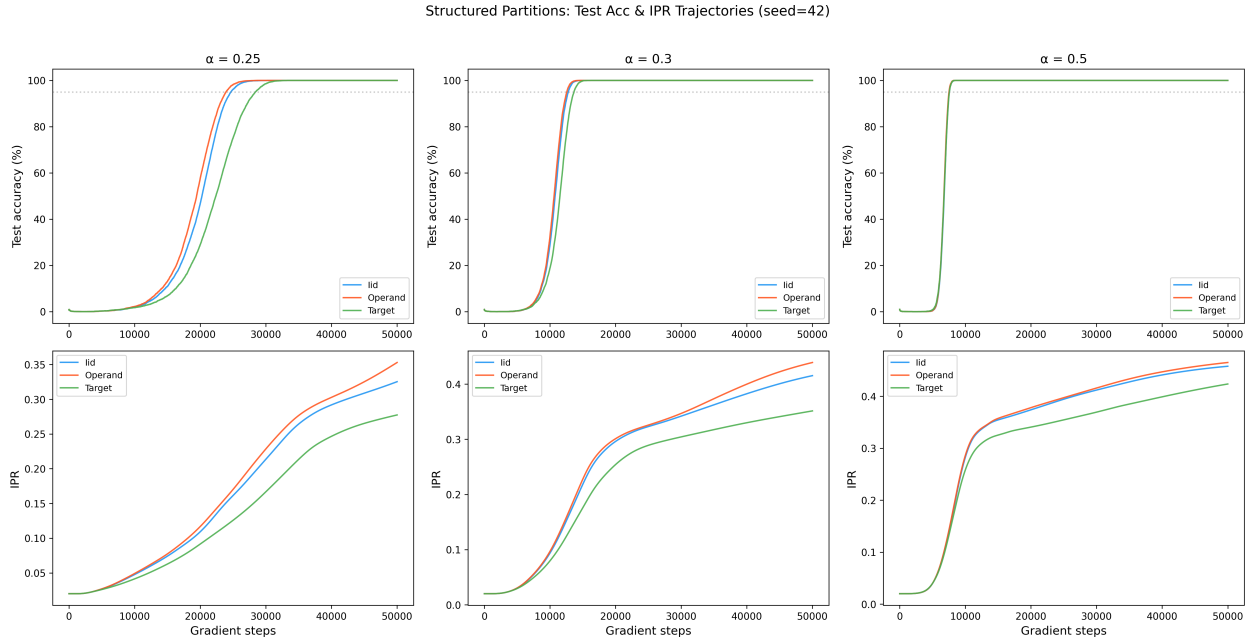


Figure 5: **Structured partition trajectories** (seed 42). Top: test accuracy; bottom: IPR. The three partition types produce nearly indistinguishable dynamics at all training fractions.

Table 2: **Exp 3b: Grokking time by partition type** ($K=10$, mean $\pm$ std over 3 seeds). Slowdown ratio is relative to IID ($\alpha_{\text{dir}}=1000$) from Exp 3a.

α	IID (T_{grok})	Operand (T_{grok})	Target (T_{grok})	Operand ratio	Target ratio
0.20	∞	∞	∞	—	—
0.25	$26,725 \pm 2,048$	$25,785 \pm 1,890$	$29,783 \pm 2,029$	$0.965\times$	$1.114\times$
0.30	$13,072 \pm 449$	$12,852 \pm 437$	$13,697 \pm 324$	$0.983\times$	$1.048\times$
0.35	$9,933 \pm 194$	$9,820 \pm 193$	$10,093 \pm 220$	$0.989\times$	$1.016\times$
0.50	$7,642 \pm 120$	$7,597 \pm 116$	$7,612 \pm 95$	$0.994\times$	$0.996\times$

1.4 Mechanistic Analysis

The summary statistics establish *what* happens (heterogeneity delays grokking near the boundary but does not prevent it). The trajectory-level diagnostic plots reveal *why*.

1.4.1 Grokking anatomy

Figure 6 overlays four metrics on a shared time axis for four representative conditions, with vertical markers at T_{onset} (test accuracy first exceeds 5%, green dotted) and T_{grok} (test accuracy permanently exceeds 95%, red dashed). The key observations:

- **Heterogeneity stretches the transition, not just the onset.** At $\alpha=0.25$, the gap between T_{onset} and T_{grok} grows from 12,085 steps (IID) to 20,645 steps ($\alpha_{\text{dir}}=0.01$)—a 71% widening. At $\alpha=0.5$, the gap is $\sim 1,800$ – $1,990$ steps regardless of heterogeneity.
- **The generalization gap peaks before T_{grok} and collapses to zero after it** (row b). Under non-IID, the peak is taller and wider, indicating that the model spends longer in the “memorized but not yet generalized” regime.
- **Client drift peaks just before T_{grok} , then decays sharply** (row c). This pattern is universal across all conditions. The peak drift under extreme non-IID is $\sim 2\times$ higher than under IID, but the post-grokking decay profile is identical—once the Fourier solution locks in, clients converge to agreement.
- **IPR and the weight norm ratio co-evolve smoothly through the transition** (row d), confirming that feature formation (IPR) and layer balancing ($\|W_2\|/\|W_1\|$) proceed in lock-step.

Table 3 quantifies the onset-to-grokking gap across all conditions. The key pattern: heterogeneity has a modest effect on T_{onset} (the onset delay is 5–17%) but a much larger effect on the transition width $T_{\text{grok}} - T_{\text{onset}}$ (up to 71% wider). The bottleneck is in the *acceleration phase*, not the ignition.

1.4.2 Weight norm dynamics

The Gromov mechanism requires weight norms to grow past a critical threshold for the periodic Fourier features to emerge. Figure 7 reveals that heterogeneity **slows weight growth in both layers**, providing a direct mechanistic explanation for the grokking delay:

- Under extreme non-IID ($\alpha_{\text{dir}}=0.01$), layer-1 and layer-2 weight norms grow ~ 20 – 30% more slowly than under IID during the pre-grokking phase.

Table 3: **Generalization onset and transition width** (seed 42). T_{onset} : first step with test accuracy $>5\%$. Gap = $T_{\text{grok}} - T_{\text{onset}}$.

α	α_{dir}	T_{onset}	T_{grok}	Gap	Gap widening
0.50	1000 (IID)	5,765	7,565	1,800	—
0.50	0.01	5,690	7,680	1,990	+11%
0.30	1000 (IID)	7,880	12,790	4,910	—
0.30	0.01	8,300	14,030	5,730	+17%
0.25	1000 (IID)	12,590	24,675	12,085	—
0.25	0.01	14,725	35,370	20,645	+71%

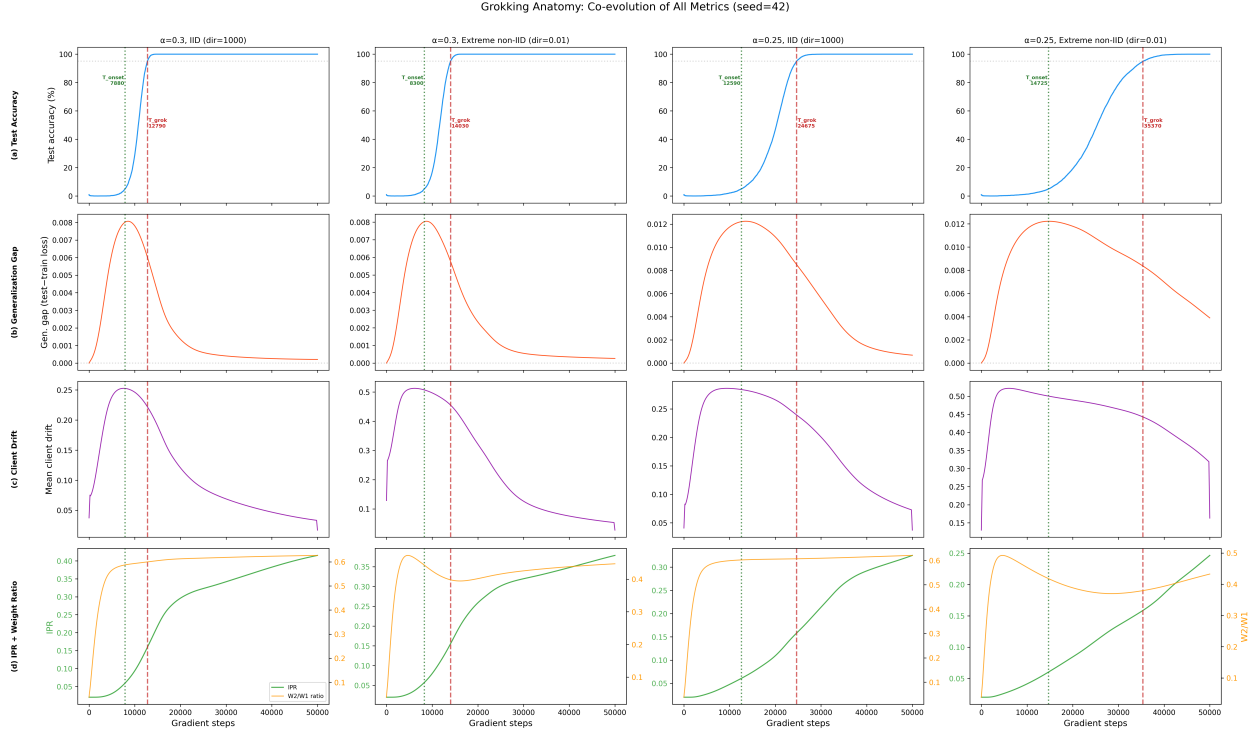


Figure 6: **Grokking anatomy** (seed 42). Four conditions spanning comfortable ($\alpha=0.3$) to near-boundary ($\alpha=0.25$), under IID and extreme non-IID. Green dotted line: T_{onset} (first $>5\%$ test accuracy); red dashed: T_{grok} ($>95\%$ permanently). Rows: (a) test accuracy, (b) generalization gap, (c) mean client drift, (d) IPR and $\|W_2\|/\|W_1\|$ ratio.

- However, the *final* weight magnitudes converge to the same values across all heterogeneity levels, confirming that the same Fourier solution is reached regardless of the data distribution.
- The weight norm ratio $\|W_2\|/\|W_1\|$ (Figure 8) converges to $\sim 0.8\text{--}0.9$ for all conditions that grok. Under non-IID, the ratio is consistently lower during training—layer 2 (the output readout) grows more slowly relative to layer 1 (the feature extractor), suggesting that aggregation noise from mismatched client label distributions differentially impedes the output layer.

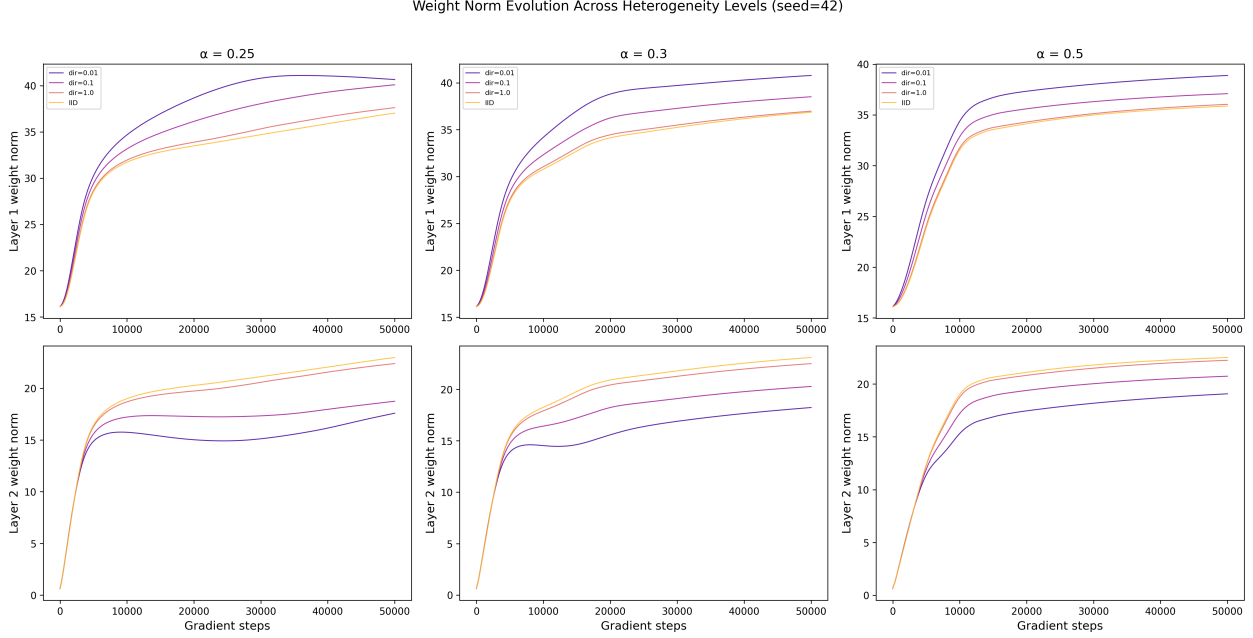


Figure 7: **Weight norm evolution** (seed 42). Top: layer-1 norm; bottom: layer-2 norm. Non-IID slows growth in both layers, with a larger effect on layer 2. Final norms converge regardless of heterogeneity.

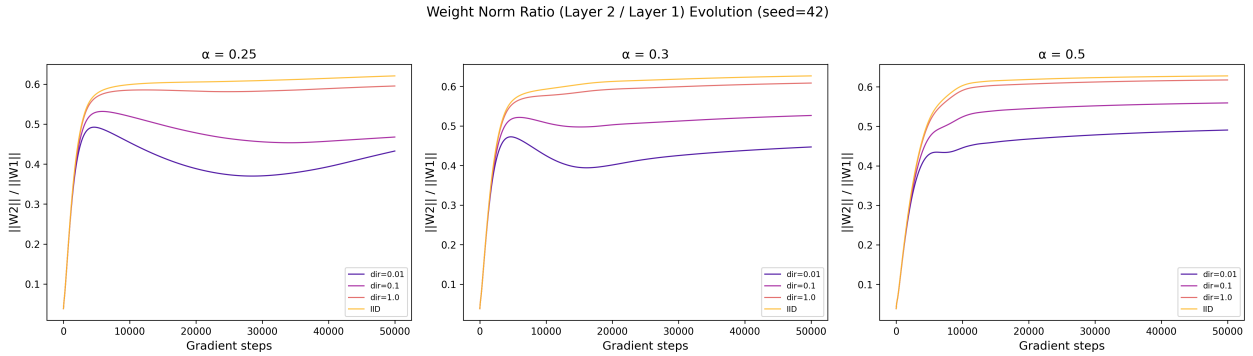


Figure 8: **Weight norm ratio** ($\|W_2\|/\|W_1\|$) over training (seed 42). All grokking runs converge to the same ratio $\approx 0.8\text{--}0.9$. Under non-IID, the ratio is suppressed during training, indicating heterogeneity preferentially slows the output layer.

1.4.3 Client drift and feature quality

Figure 9 shows mean client drift and client weight divergence across heterogeneity levels. Two distinct phenomena are visible:

- **Drift magnitude scales with heterogeneity** throughout training (top row): $\alpha_{\text{dir}}=0.01$ produces 3–5 $\times$ higher drift than IID.
- **Weight divergence spikes at the grokking transition** (bottom row). This spike appears in all conditions and coincides precisely with the test accuracy jump. It suggests that grokking is a cooperative phase transition that temporarily destabilizes client consensus—clients briefly explore different regions of the solution landscape before snapping into agreement on the Fourier solution.

Despite high drift, Figure 10 shows that **feature quality at grokking is independent of drift level**. The IPR at T_{grok} clusters tightly around 0.13–0.16 for all grokked runs (across all α , α_{dir} , and seeds), with no correlation to the mean client drift experienced during training. Drift delays grokking but does not degrade the solution.

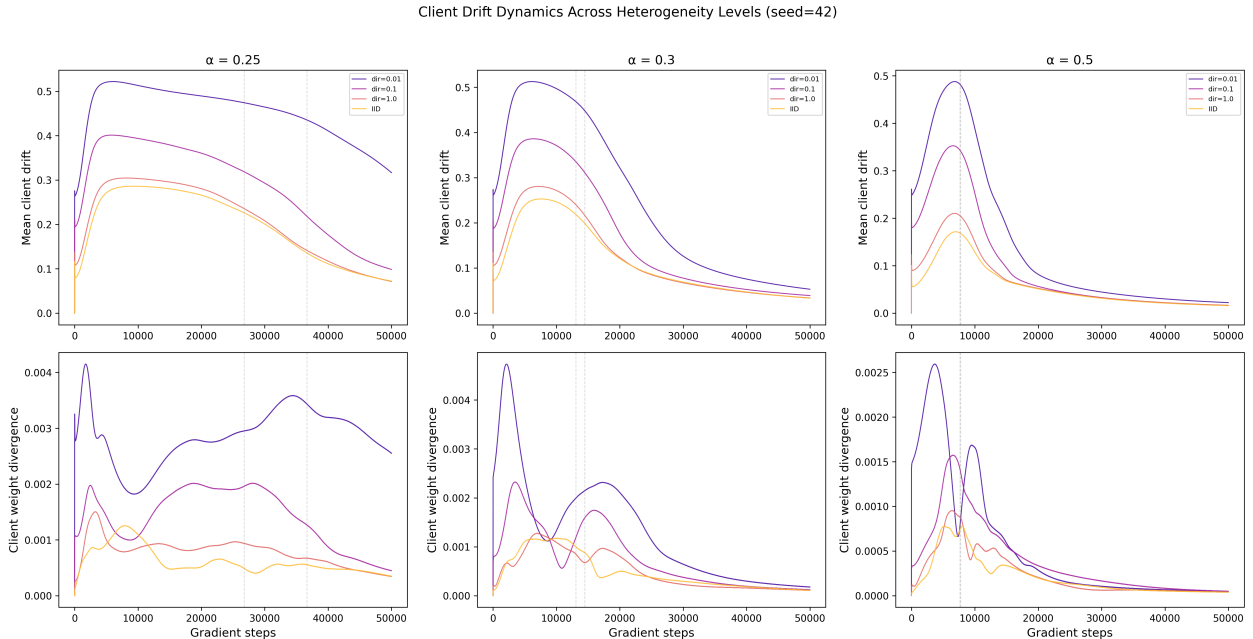


Figure 9: **Client drift dynamics** (seed 42). Top: mean client drift; bottom: client weight divergence. Drift scales with heterogeneity; weight divergence shows a characteristic spike at the grokking transition.

1.4.4 Reproducibility and seed variability

Figure 11 shows per-seed trajectories for four conditions spanning comfortable and near-boundary regimes under IID and extreme non-IID:

- At $\alpha=0.5$, all three seeds produce tightly clustered trajectories (T_{grok} varies by $\sim 1,000$ steps) regardless of heterogeneity.

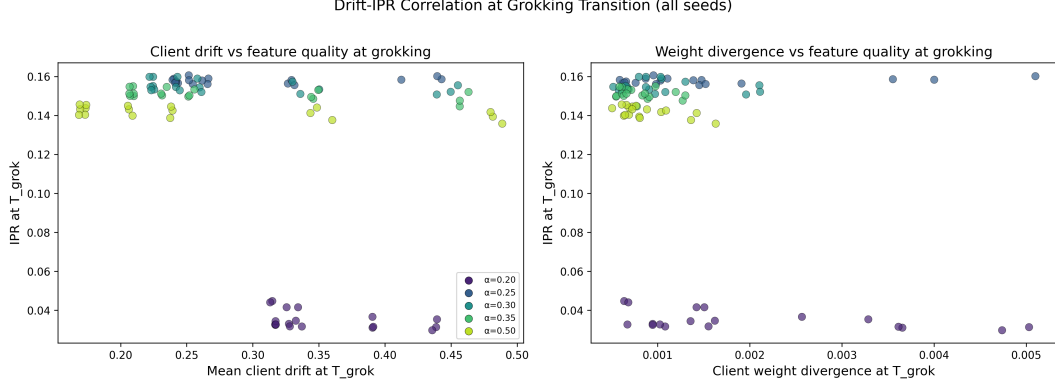


Figure 10: **Drift–IPR correlation at grokking.** Each point is one run (all seeds). Left: client drift vs. IPR at T_{grok} ; right: weight divergence vs. IPR. Among grokked runs, IPR is tightly clustered regardless of drift—feature quality is independent of the heterogeneity experienced during training. The low-IPR cluster ($\sim 0.03\text{--}0.06$) corresponds to $\alpha=0.2$ runs that did not grok.

- At $\alpha=0.25$ with $\alpha_{\text{dir}}=0.01$ (the most stressed condition that still groks), seed variance increases substantially: the fastest seed groks at $\sim 35,000$ steps while the slowest requires $\sim 39,000$. The grokking transition becomes a more stochastic event near the phase boundary under high heterogeneity.
- Despite timing variability, all seeds reach 100% test accuracy and converge to the same final IPR (~ 0.39), confirming that the outcome is deterministic even when the timing is not.

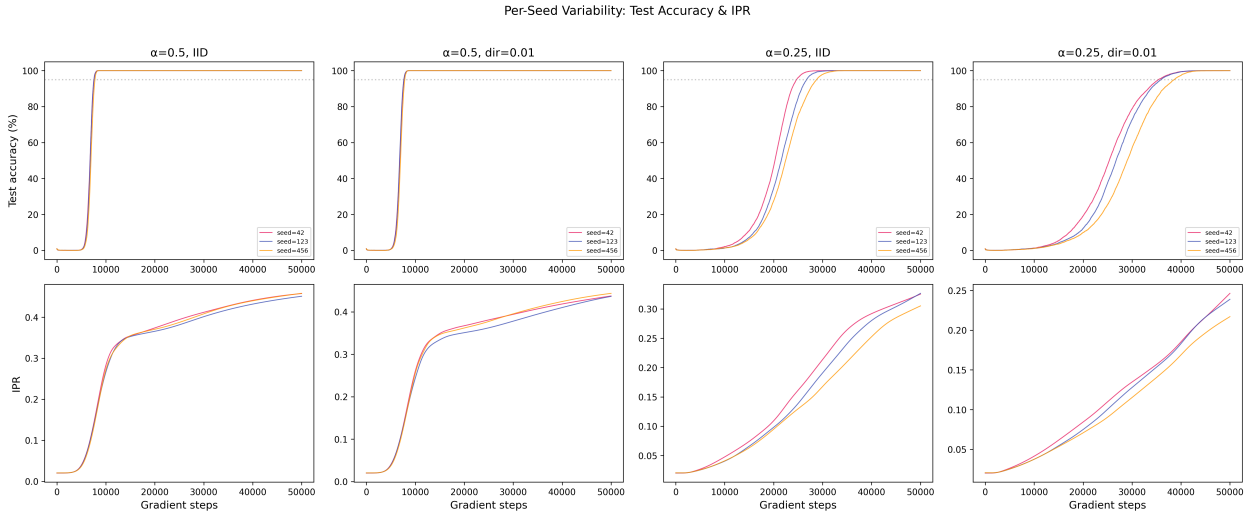


Figure 11: **Per-seed variability.** Top: test accuracy; bottom: IPR. Columns span from comfortable ($\alpha=0.5$, IID) to maximally stressed ($\alpha=0.25$, $\alpha_{\text{dir}}=0.01$). Timing variance increases near the boundary under heterogeneity, but all seeds converge to the same solution.

1.5 Summary of Findings

The results from Experiment 3 support a clear narrative:

1. **Grokking is robust to heterogeneity.** Across 147 federated runs spanning Dirichlet, operand, and target partitions, *every* run with $\alpha \geq 0.25$ reached 100% test accuracy. The phase boundary $\alpha_{\text{crit}} \approx 0.2$ is unchanged by heterogeneity.
2. **Heterogeneity delays but does not prevent grokking.** The delay is concentrated near the phase boundary: up to 37% slowdown at $\alpha=0.25$ under extreme non-IID ($\alpha_{\text{dir}}=0.01$), but $<1.3\%$ at $\alpha=0.5$.
3. **The delay mechanism is weight growth retardation.** Non-IID data slows the growth of weight norms in both layers, particularly layer 2, delaying the point at which the Fourier features become sufficiently strong to drive generalization.
4. **Structured partitions do not fragment Fourier learning.** Contrary to the initial hypothesis, operand-based partitioning (which fragments the input structure) produces grokking times within noise of IID. The Fourier solution is learned through aggregation regardless of how the input space is divided across clients.
5. **Client drift does not degrade the final solution.** High drift during training delays grokking but has no effect on the quality of the learned features (IPR at convergence is identical across all conditions).
6. **The grokking transition is a cooperative phase transition.** The characteristic spike in client weight divergence at T_{grok} , the universal drift-peak-then-decay pattern, and the lockstep co-evolution of IPR and weight norm ratios all point to grokking as a sharp, coordinated transition in parameter space that occurs simultaneously across all clients despite their heterogeneous local data.